

Assignment 1. Working with Directories and Files in Python

Deadline: Upload the assignment to Moodle and defend it by **September 25, 2026**.

If the assignment is uploaded and defended after the deadline, **15% will be deducted from the total score**. Therefore, even if all tasks are completed correctly, the maximum possible score will be **85 points**.

If the report is uploaded and defended after the deadline, **25% will be deducted from the total score**. Therefore, even if all requirements are completed correctly, the maximum possible score will be **75 points**.

Task 1: (25 points) Working with Folders and Files

1. Creating the project structure

- Write a script that creates a project directory with the following subfolders:
 - data
 - results

2. Working with CSV files

- In the data folder, save a students.csv file containing student data (full name, group, grades).
- Read the data from students.csv and calculate the average grade for each student.

3. Saving the results

- Save the aggregated results in JSON format (results/report.json).
- Check whether the file already exists. If it does, display a warning and ask for confirmation before overwriting it.

4. Archiving

- Automatically create a results.zip archive containing the results folder.

5. Working with pathlib

- Check whether report.json exists. If it does, print its size and last modification date.

Task 2: (25 points) Search for a Word or Phrase in Text Files

Generate five text files (name_.txt) and place them in the same folder. The files may contain any text. Write a program that accepts a search query and outputs the names of the text files containing the specified substring.

Input format

A string containing the search query. It may be a single word or a phrase. For example: mathematics, probability theory.

Output format

A list of text files containing the substring entered by the user, or a message stating that the search query is not found in these files.

Task 3: (25 points) File Information

A file named `file.txt` is provided with the assignment. Write a program that outputs the following text statistics:

- the number of letters from the Latin alphabet;
- the number of words;
- the number of lines.

Task 4: (25 points) Replacing Words

Two files, `words.txt` and `forbidden_words.txt`, are provided with the assignment. The file `words.txt` contains text. The file `forbidden_words.txt` contains forbidden words separated by spaces.

Write a program that replaces all forbidden words from `forbidden_words.txt` in `words.txt` with asterisks `*` (the number of asterisks must equal the number of letters in the word).

The program must replace all forbidden words wherever they occur, even in the middle of another word. Replacement is case-insensitive: if `forbidden_words.txt` contains the word `exam`, then `exam`, `Exam`, `ExaM`, `EXAM`, and `exAm` must all be replaced with `****`.

Output format

Text edited according to the task requirements.

Self-Check Questions

1. How can you open a file for reading and writing in Python?
2. What is the difference between the file opening modes `'r'`, `'w'`, `'a'`, `'rb'`, and `'wb'`?
3. What is the advantage of using the `with open(...)` as `f:` construct when working with files?
4. How should a file be closed correctly, and why is this important?
5. What are the main functions of the `os` module used for working with directories?
6. How can you create a new folder or delete an existing one in Python?
7. What is the difference between `os.getcwd()` and `os.chdir()`?
8. How can you get a list of files and folders in a directory?
9. What is the `os.path` module used for?
10. How can you check whether a file or folder exists?
11. Why is `pathlib` more convenient than `os.path`?
12. How can you combine file paths in a cross-platform way?
13. What Python methods can be used to read a CSV file?
14. How is the `csv` module useful compared with ordinary line-by-line file reading?
15. How can Python data be serialized to JSON format?
16. What is the difference between working with JSON and CSV?
17. Which Python data types can be stored directly in JSON?
18. What is lazy loading, and why is it useful?

19. How can you read a large file line by line without loading the entire file into memory?
20. What is the purpose of the `readline()` method and of iterating over a file object?
21. When should generators be used when working with files?
22. Which Python modules can be used to work with archives (.zip, .tar)?
23. How can you create a ZIP archive in Python?
24. How can you extract an archive to a specified directory?
25. What are the key considerations when working with binary files (for example, .png or .exe)?